

Aim:

To obtain the VI characteristics of the semiconductor diode for the

- * PN Junction diode
- * Point-contact diode
- * Zener diode

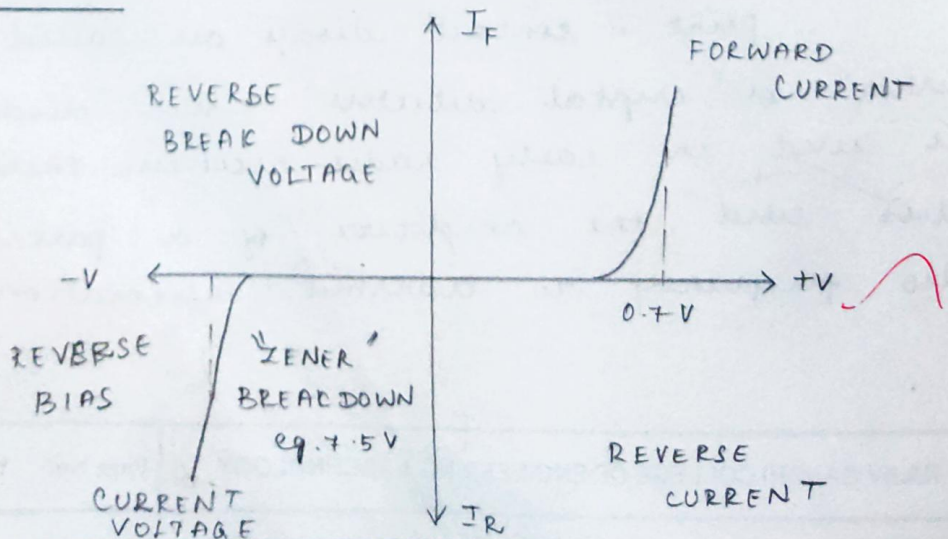
Apparatus required:

SL No.	Apparatus Name	Type / Range	Quantity
1.	Regulated power supply (RPS)	(0-30) V	1
2.	Voltmeter (V_1)	(0-2) V	1
3.	Voltmeter (V_2)	(0-10) V	1
4.	Ammeter (I_1)	(0-50) mA	1
5.	Ammeter (I_2)	(0-100) μ A	1
6.	Bread Board	-	1

C. ZENER DIODE

FORWARD BIAS		REVERSE BIAS	
$V_F (V)$	$I_F (mA)$	$V_R (V)$	$I_R (mA)$
0.14	0.00	0.14	0.01
0.28	0.00	0.41	0.04
0.52	0.00	0.61	0.06
0.67	0.07	0.82	0.08
0.70	0.27	0.82	0.09
0.72	0.37	0.97	0.12
0.73	0.66	1.23	0.13
0.74	0.84	1.39	0.16
0.75	0.99	1.63	0.18
0.76	1.23	1.83	0.20
0.76	1.38	2.03	0.21
0.77	1.63	2.14	0.24
0.79	1.66	2.34	0.33
0.81	2.07	2.57	0.39
0.82	2.15	2.83	0.51
0.83	2.43	2.97	0.73
0.84	2.57	3.18	1.12
0.84	2.74	3.39	1.81
0.86	3.03	3.59	2.67
0.89	3.12		

MODEL GRAPH:

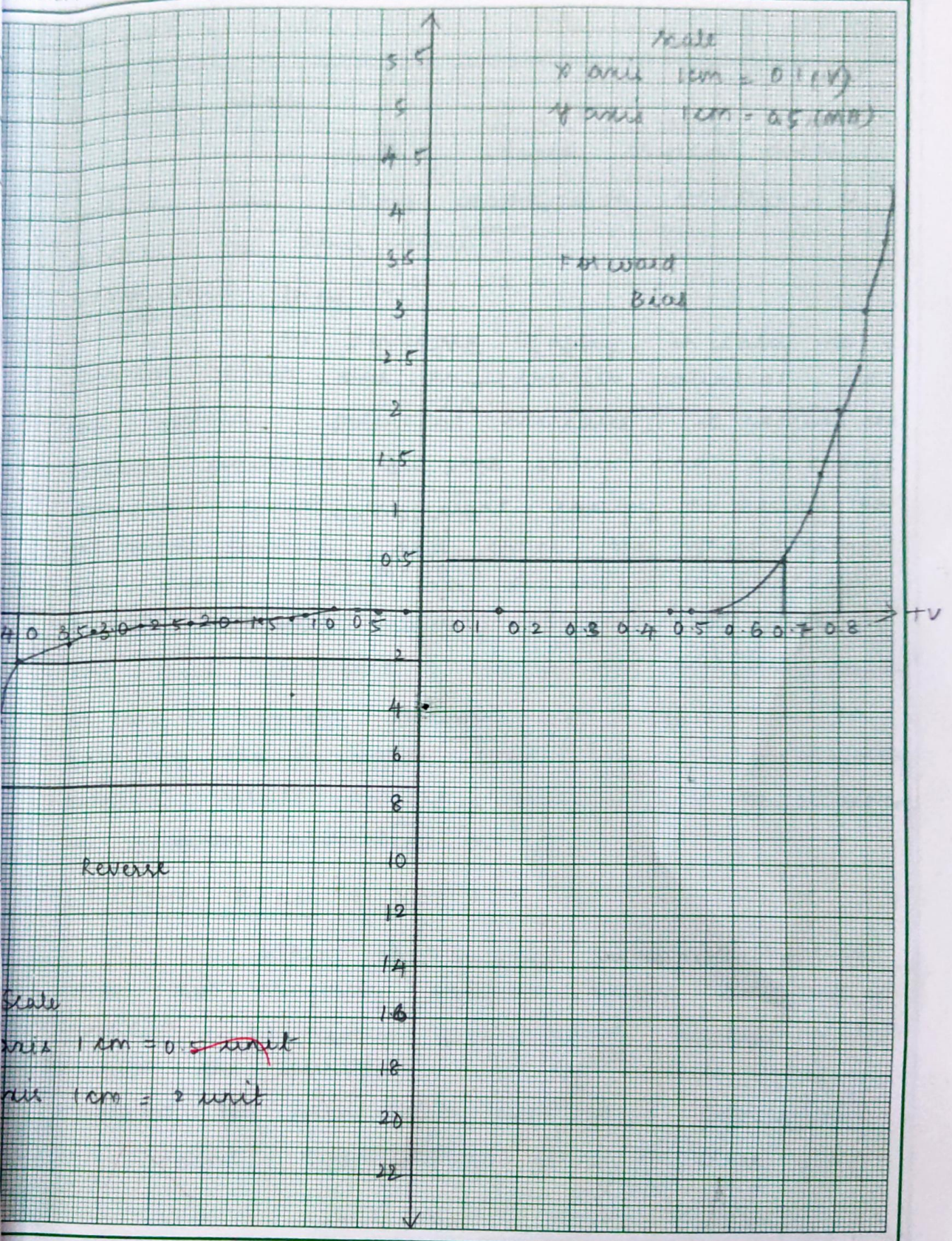


Zener diode

I_F

Date :

atch No:



I_{Z1}

In FB,

$$\text{static } R_S = \frac{V_F}{R_F}$$

$$= \frac{0.8}{2}$$

$$= 0.4 \times 10^3 \Omega$$

$$R_d = \frac{\Delta V_F}{\Delta I_F}$$

$$= \frac{0.8 - 0.7}{2 - 0.5 \times 10^{-3}}$$

$$= 0.06 \times 10^3 \Omega$$

In RB

$$\text{static } R_S = \frac{V_F}{I_F}$$

$$= \frac{4.25}{7 \times 10^{-6}}$$

$$= 0.64 \times 10^6 \Omega$$

$$\text{Dynamic } R_d = \frac{\Delta V_F}{\Delta I_F}$$

$$= \frac{7 - 2}{4.25 - 4.0}$$

$$= \frac{5}{0.25 \times 10^{-6}}$$

$$= 20 \times 10^6 \Omega$$

Procedure:

Forward Bias:

- * Connect the circuit as per the circuit diagram.
- * Vary the power supply in such a way in that readings are taken.
- * Notedown the corresponding ammeter readings.
- * plot the graph V_F & I_F .
- * Find the dynamic resistance.

Reverse Bias:

- * Connect the circuit as per the circuit diagram.
- * Vary the power supply in such a way that the readings are taken in the steps.
- * Note down the corresponding ammeter reading I_R .
- * plot the graph between V_R and I_R .
- * Find the dynamic resistance.



RESULT:

Subject

Thus V-I characteristics of the PN Junction diode, point contact diode and Zener diode are discussed and their static and dynamic resistance obtained are result below.

	Forward Bias		Reverse Bias	
	Static	Dynamic	Static	Dynamic
P-N Junction diode	$0.15 \times 10^3 \Omega$	$0.66 \times 10^3 \Omega$	$0.1 \times 10^6 \Omega$	$0.1 \times 10^6 \Omega$
point contact diode	$0.4 \times 10^3 \Omega$	$0.66 \times 10^3 \Omega$	$0.60 \times 10^6 \Omega$	$20 \times 10^6 \Omega$
Zener diode	$0.4 \times 10^3 \Omega$	$0.66 \times 10^3 \Omega$	$0.64 \times 10^6 \Omega$	$20 \times 10^6 \Omega$